

Started on	Friday, 22 May 2026, 9:13 PM
State	Finished
Completed on	Friday, 22 May 2026, 9:17 PM
Time taken	4 mins 2 secs
Grade	100.00 out of 100.00

Question 1

Correct

Mark 50.00 out of 50.00

Use [LU Decomposition](#) to find L and U matrix.

For example:

Input	Result
[[3, 2, 7], [2, 3, 1], [3, 4, 1]]	<pre>[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]</pre>

Answer: (penalty regime: 0 %)

Reset answer

```
1 '''Program to find L and U matrix using LU decomposition.
2 Developed by: THENAMUDHEN S
3 RegisterNumber: 212225040471
4 '''
5 import os
6 os.environ["OPENBLAS_NUM_THREADS"]="1"
7 import numpy as np
8 from scipy.linalg import lu
9 matrix = np.array(eval(input()))
10 P,L,U=lu(matrix)
11 print(L)
12 print(U)
```

	Input	Expected	Got	
✓	[[3, 2, 7], [2, 3, 1], [3, 4, 1]]	<pre>[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]</pre>	<pre>[[1. 0. 0.] [1. 1. 0.] [0.66666667 0.83333333 1.]] [[3. 2. 7.] [0. 2. -6.] [0. 0. 1.33333333]]</pre>	✓
✓	[[5, 1, 8], [4, 5, 7], [8, 9, 1]]	<pre>[[1. 0. 0.] [0.625 1. 0.] [0.5 -0.10810811 1.]] [[8. 9. 1.] [0. -4.625 7.375] [0. 0. 7.2972973]]</pre>	<pre>[[1. 0. 0.] [0.625 1. 0.] [0.5 -0.10810811 1.]] [[8. 9. 1.] [0. -4.625 7.375] [0. 0. 7.2972973]]</pre>	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 50.00/50.00.

Question **2**

Correct

Mark 50.00 out of 50.00

Use [LU Decomposition](#) to solve a matrix.**For example:**

Input	Result
[[3, 2, 7], [2, 3, 1], [3, 4, 1]] [4, 5, 7]	[0.875 1.125 -0.125]

Answer: (penalty regime: 0 %)

Reset answer

```

1  '''Program to solve a matrix using LU decomposition.
2  Developed by: THENAMUDHEN S
3  RegisterNumber: 212225040471
4  '''
5  import os
6  os.environ["OPENBLAS_NUM_THREADS"]="1"
7  import numpy as np
8  from scipy.linalg import lu_factor,lu_solve
9  matrix= np.array(eval(input()))
10 constant= np.array(eval(input()))
11 piv,lu= lu_factor(matrix)
12 result= lu_solve((piv,lu),constant)
13 print(result)

```

	Input	Expected	Got	
✓	[[3, 2, 7], [2, 3, 1], [3, 4, 1]] [4, 5, 7]	[0.875 1.125 -0.125]	[0.875 1.125 -0.125]	✓

Passed all tests! ✓

► Show/hide question author's solution (Python3)

Correct

Marks for this submission: 50.00/50.00.